

0.1 Notes on Checksum Communication Complexity

A paper has shown that the Number on Forehead communication complexity of CheckSum has been bounded as follows:

$$(\log N)^{\Omega(1)} \leq NOF(CheckSum) \leq O(\sqrt{\log N})$$

There are many different techniques to analyze something called Ramsey type Problems, where we try to maximize the size of a set while avoiding a certain pattern (like 3-Arithmetic Progressions).

- Graph Theory
- Ergodic Theory
- Fourier Analysis (Most Success)
- Polynomial Method

1 3AP Over Finite Field

The 3AP problem is the same as the original problem over $[N]$, however, we focus on the case where arithmetic is done over $\mathbb{R}_3^n$. This argument generalizes to any group where there exists a plus operation and we make a set such that:

$$\nexists a, b, c | a \neq b \neq c, a + c = 2b$$

We want to determine, given the universe $U = \{0, 1, 2\}^n$ where addition is mod 3:

$$r_3(\mathbb{F}_3^n) = \max |S| \text{ where } S \subseteq \{0, 1, 2\}^n \text{ and no 3AP exists.}$$

This is also called the "CAP-SET PROBLEM".

1.1 Size of CAP-SET

Recall that Behrend's construction led to a subset $S \subseteq [N]$, where:

$$|S| \subseteq \frac{N}{2^{c \cdot \sqrt{\log N}}}$$

In CAPSET case, the size of N is 3^n . We argue that it is easier to form a 3AP in $\mathbb{F}_3^n$ than in $[3^n]$.

Intuition: Finding solutions to $x + y = 2z$ modular arithmetic is easier over $\mathbb{F}_3^n$ than $\mathbb{Z}$. (I do not get his argument)

Theorem 1.1. $r_3(\mathbb{F}_3^n) \leq (2.76)^n = N^c$ for some $c < 1$.

Recall that $N/(2^{c\sqrt{\log N}}) \leq r_3(\mathbb{Z}_n)$

1.2 Examples and Properties

Firstly, we know that we can select a set that has size 2^n that satisfies that there are no 3AP's. We argue that $A = \{0, 1\}^n$ has no 3AP's.

Proof. Assume that there was some a, b, c such that they form a nontrivial 3AP. The argument is relatively straightforward, but we know that if $a_i + b_i$ is 1, c_i cannot be a three AP, if $a_i + b_i = 0$ then $a_i = b_i = c_i = 0$ so they must be the same bit and if $a_i + b_i = 2$ then $c_i = b_i = a_i = 1$. So either the three numbers are the same or they do not form a 3AP. $\square$

This provides a lower bound for $r_3(\mathbb{F}_3^n)$:

$$r_3(\mathbb{F}_3^n) \geq 2^n = N^{\log_3 2} \approx N^{0.63}$$

What do 3APs over mod 3 arithmetic look like? We know that:

$$(a, b, c) \in \{0, 1, 2\}^n \text{ and } a + b = 2c$$

We can think of it as:

$$\begin{aligned} a + b &\equiv 2c \pmod{3} \\ \forall i, a_i + b_i &\equiv 2c_i \pmod{3} \\ a_i + b_i &\equiv -c_i \pmod{3} \\ a_i + b_i + c_i &\equiv 0 \pmod{3} \end{aligned}$$

This equation is satisfied when all $a_i = b_i = c_i$ or each value is unique.

1.3 Lower Bound on $r_3(\mathbb{F}_3^n)$

Theorem 1.2. $r_3(\mathbb{F}_3^n) \leq (2.76)^n = N^c$ for some $c < 1$.

Proof Outline: We are going to use the polynomial method which will aim to do the following:

- Encode objects or desired solutions as polynomials
- Use linear algebra (rank / dimension)

- Use interpolation
- Fast polynomial evaluation

Generally we will show that if there exists some large 3AP free set, it implies that some property of identity tensor is violated.

Definition 1.1. We say that a **Rank** of a matrix M ($rank(M)$) is the number of linearly independent columns or number of linearly independent rows of the matrix. Another way to think about it is the number of rank 1 matrices that add up to M :

$$Rank(M) = \min_r \{M = M_1 + \dots + M_r\}$$

Where all M_i are rank 1 matrices.

Knowing that $rank(M) = r$, we can write M as the following:

$$M = f_1 g_1^T + \dots + f_r g_r^T$$

Where each f_i represents one of the linearly independent columns and each g_i represents one of the linearly independent rows. This decomposition has each $M_i = f_i \cdot g_i^T$. We can therefore redefine rank for matrices in $\mathbb{F}_3^{n \times n}$ as the following:

$$Rank(M) = \min_r \left\{ \begin{array}{l} \exists f_1, \dots, f_r : [N] \longrightarrow \mathbb{F}_3, \\ \exists g_1, \dots, g_r : [N] \longrightarrow \mathbb{F}_3 \\ | M(x, y) = f_1(x)g_1(y) + \dots + f_r(x)g_r(y) \quad \forall x, y \in [N] \end{array} \right\}$$

We should also think of matrices as a function that takes in two inputs and outputs a value. Furthermore, for a tensor, we can think of it as a function that takes in three inputs and outputs a value. For tensor rank, the general notion is as follows:

Definition 1.2. We say that a **Rank** of a tensor T ($rank(T)$) is the number of rank 1 tensors that add up to T :

$$Rank(T) = \min_r \{T = T_1 + \dots + T_r\}$$

Where all T_i are rank 1 tensors.

We draw a parallel with the second definition of matrix rank and define the following for tensors:

$$\begin{aligned} \text{Rank}(T) = \min_r \Big\{ & \exists f_1, \dots, f_r : [N] \longrightarrow \mathbb{F}_3, \\ & \exists g_1, \dots, g_r : [N] \longrightarrow \mathbb{F}_3, \\ & \exists h_1, \dots, h_r : [N] \longrightarrow \mathbb{F}_3, \\ & | T(x, y, z) = f_1(x)g_1(y)h_1(z) + \dots + f_r(x)g_r(y)h_r(z) \quad \forall x, y, z \in [N] \Big\} \end{aligned}$$

You think of a rank one tensor as a function that can be written as a product of three functions, each of which takes in one of the three inputs. We will be interested in the following tensor:

$$\mathbb{I}(x, y, z) = \begin{cases} 1 & \text{if } x = y = z \\ 0 & \text{otherwise} \end{cases}$$

Firstly, we know that $\text{rank}(\mathbb{I}^{N \times N}) \leq N$ because we can write $\mathbb{I}$ as the sum of N rank one tensors. Secondly, we know that $\text{rank}(\mathbb{I}^{N \times N \times N}) \leq N$ where $f_i(x) = g_i(y) = h_i(z) = 1$ if $x = y = z = i$ and 0 otherwise. For each f_i, g_i, h_i we just say $f_i(x) = 1$ if $x = i$ and 0 otherwise.

1.3.1 Thought Experiment

Say we have a matrix M which we think about as a function $M(x, y)$. If two parties are trying to figure out the value of $M[x, y]$ where one party has x and the other has y , one easy protocol is having the first party send x to the second party and having the second party compute $M[x, y]$. However, if M has low rank, we can do better. We can write M as the sum of r rank one matrices:

$$M = f_1 g_1^T + \dots + f_r g_r^T$$

In this case, we can have Alice (party 1) send $f_i(x)$ for all i to Bob. Since $f_i(x)$ is either $[0, 1, 2]$, the complexity of this protocol is just $O(r)$ opposed to $O(\log N)$.

Consider the case with three parties trying to compute the tensor $M^{n \times n \times n}$. If M has low rank, we can write M as the sum of r rank one tensors:

$$M = f_1 g_1 h_1 + \dots + f_r g_r h_r$$

In this case, we can have Alice and Bob send $f_i(x)$ and $g_i(y)$ for all i to Charlie, and Charlie can compute $M[x, y, z]$ by computing $f_i(x)g_i(y)h_i(z)$ for all i . The complexity of this protocol is just $O(r)$ opposed to $O(\log N)$ once more.

1.3.2 New Notion of Tensor Rank

We define a new notion of tensor rank called Slice Rank. To motivate this, we make a continuation of the previous thought experiment where each round, two players can get together and send information to the third player. We define the protocol as:

- Alice has x , Bob has y , and Charlie has z .
- Start with Alice and Bob in a room and Charlie in another room. Alice and Bob can talk to each other but not to Charlie.
- In other rounds, any two players can get together and talk to each other but not to the third player.

To formalize this idea, we define the following:

Definition 1.3. We say that a **Slice Rank** of a tensor T ($srank(T)$)

$$\text{SRank}(T) = \min \left\{ r : \begin{array}{l} \exists r_1, r_2, r_3 \text{ with } r_1 + r_2 + r_3 = r, \\ f_1, \dots, f_{r_1} : [M] \rightarrow \mathbb{F}, \\ \tilde{f}_1, \dots, \tilde{f}_{r_1} : [M] \times [M] \rightarrow \mathbb{F}, \\ g_1, \dots, g_{r_2} : [M] \rightarrow \mathbb{F}, \\ \tilde{g}_1, \dots, \tilde{g}_{r_2} : [M] \times [M] \rightarrow \mathbb{F}, \\ h_1, \dots, h_{r_3} : [M] \rightarrow \mathbb{F}, \\ \tilde{h}_1, \dots, \tilde{h}_{r_3} : [M] \times [M] \rightarrow \mathbb{F}, \\ \text{such that} \\ T(x, y, z) = \sum_{i=1}^{r_1} f_i(x) \tilde{f}_i(y, z) + \sum_{i=1}^{r_2} g_i(y) \tilde{g}_i(x, z) \\ \quad \quad \quad + \sum_{i=1}^{r_3} h_i(z) \tilde{h}_i(x, y) \end{array} \right\}.$$

We end this lecture by explaining that $SRank(T) \leq Rank(T)$ since we can just define $\tilde{f}_i$ as $g_i \cdot h_i$.

2 Lecture 9

First, we define a tensor associated with $3AP$ sets, and show that if there exists a large $3AP$ free set, some property of the tensor is violated. Consider

the following function for checksum:

$$CS : \mathbb{F}_3^n \times \mathbb{F}_3^n \rightarrow \mathbb{F}_3 \rightarrow \{0, 1\} \quad (1)$$

$$CS(x, y, z) = \begin{cases} 1 & \text{if } x + y + z = 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

For this function, we can think of it like a cube with 1's along the diagonal, and 0's in other locations unless the three inputs are all different, or all sum to 0 mod 3. Consider a subset $S \subseteq \mathbb{F}_3^n$ such that S is a $3AP$ free set, meaning no $x + y + z = 0$ for distinct $x, y, z \in S$. This implies that $CS(x, y, z) = 1$ if and only if $x = y = z$ (We think of the subset S as taking a subcube of the tensor by removing the i th element row). Because of this, the remaining tensor representing S must be the identity tensor. In otherwords:

$$CS_{S \times S \times S} = \mathbb{I}^{S \times S \times S}$$

2.1 Properties of Identity Tensor

We will prove the three following properties:

1. $SRank(\mathbb{I}_{M \times M \times M}) = M$
2. $SRank(\text{ sub-tensors }) \leq SRank(\text{ main tensor })$
3. $SRank(CheckSum) \leq 3^{cn}$ where $(c < 1)$

This is sufficient to demonstrate that there does not exist a big S . Since $CS_{S \times S \times S} = |S|$ from property 1. Then we know that $|S| \leq SRank(CS)$ by 2. If we bound $SRank(CS)$, we therefore have a bound for $|S|$.

2.1.1 Bounding Slice Rank of Identity Tensor

We aim to show that the slice rank of the identity tensor is M . Firstly, we know that for matrices we can easily show that I_M has rank M since there are M linearly independent rows/columns. We, however, cannot extend this argument to Tensors so we try another approach.

Lemma 2.1. $SRank(\mathbb{I}_{M \times M \times M}) = M$

First, consider the following tensor $T : [M] \times [M] \times [M] \rightarrow \{0, 1\}$

$$T(x, y, z) = \begin{cases} 1 & \text{if } x + y + z \text{ is even} \\ 0 & \text{otherwise} \end{cases}$$

The above has a $SRank$ of 2:

$$T(x, y, z) = \mathbb{1}(x \text{ is even}) \cdot \mathbb{1}(y + z \text{ is even}) + \mathbb{1}(x \text{ is odd}) \cdot \mathbb{1}(y + z \text{ is odd})$$

We cannot extend proofs that $I_{M \times M}$ has rank M that rely on linear independence or eigenvalues. To come up with a proof that extends to tensors, we use the decomposition definition where we can rewrite the matrix as:

$$I_d = \sum_i^{M-1} f_i(x) \cdot g_i(y)$$

Recall that we can think of each $f_i(x) \cdot g_i(y)$ like two vectors $f_i \cdot g_i^\top$. Therefore, there must exist a function $u : [M] \rightarrow \mathbb{R}$ that is perpendicular to all $f_1, f_2, \dots, f_{M-1}$. Therefore, we know that:

$$\begin{aligned} u^\top \mathbb{I}_d &= u^\top \left(\sum_i^{M-1} f_i \cdot g_i^\top \right) \\ &= \sum_i^{M-1} (u^\top \cdot f_i) \cdot g_i^\top \\ &= 0 \end{aligned}$$

However, this is a contradiction since the result should be $u^\top$. We extend this to tensors.

First we show that $SRank(\mathbb{I}_{M \times M \times M}) \leq M$. This is because we can rewrite the tensor as follows:

$$\mathbb{I}_{M \times M \times M} = \sum_{a=1}^M \mathbb{1}(x = a) \cdot \mathbb{1}(y = a) \cdot \mathbb{1}(z = a)$$

Now we show that $SRank(\mathbb{I}_{M \times M \times M}) \geq M$:

Proof. Suppose $SRank(I) < M$. Then we can write it as follows:

$$\begin{aligned} I(x, y, z) &= \sum_{i=1}^{r_1} f_i(x) \tilde{f}_i(y, z) + \sum_{i=1}^{r_2} g_i(y) \tilde{g}_i(x, z) \\ &\quad + \sum_{i=1}^{r_3} h_i(z) \tilde{h}_i(x, y) \end{aligned}$$

Where $r_1 + r_2 + r_3 = r$. First, we find a new "canceling" function F such that $\langle f_i, F \rangle = 0$ for $i = 1, \dots, r_1$ and F has at least $M - r_1$ non-zero values. We argue that such an F must always exist. This is because. First it follows that if a set of vectors has rank r_1 where $r_1 < M$, then there must be a function/vector that is perpendicular to it. We argue that one of these vectors must have at least $M - r_1$ nonzero entries. Note that the span of all vectors f_i is at most r_1 and the span of all vectors not in the span is $M - r_1$. Since the perpendicular vectors span $M - r_1$, you must be able to find a vector with that many non-zero entries.

We define a new matrix:

$$\begin{aligned} G(y, z) &= \sum_{x \in [M]} I(x, y, z) \cdot F(x) \\ &= \sum_{x \in [M]} \left(\sum_{i=1}^{r_1} f_i(x) \tilde{f}_i(y, z) + \sum_{i=1}^{r_2} g_i(y) \tilde{g}_i(x, z) \right. \\ &\quad \left. + \sum_{i=1}^{r_3} h_i(z) \tilde{h}_i(x, y) \right) F(x) \end{aligned}$$

This is the matrix taken when we sum over all x values. It is essentially the analog of the $u^\top \cdot \mathbb{I}_d$ from the matrix example. The resulting function is over just y and z , giving us:

$$\begin{aligned} G(y, z) &= \sum_{x \in [M]} \mathbb{I}_d(x, y, z) \cdot F(x) \\ &= \sum_{i=1}^{r_2} g_i(y) \left(\sum_{x \in [M]} \tilde{g}_i(x, z) \cdot F(x) \right) + \sum_{i=1}^{r_3} h_i(z) \left(\sum_{x \in [M]} \tilde{h}_i(x, y) \cdot F(x) \right) \end{aligned}$$

We can visualize this new matrix as follows:

$$\begin{pmatrix} F(1) & 0 & \dots & 0 \\ 0 & F(2) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & F(n) \end{pmatrix}$$

This is because the original tensor is one ONLY when $x = y = z$ and a 0 everywhere else. So therefore, taking the summation over all $x \in [M]$ should yield a 1 only once in the new matrix, when the y and z values are equal to it. These 1's are then multiplied by $F(x)$ which results in the diagonal being $F(x)$.

As stated earlier, there are at least $M - r_1$ values of $F(x)$ that are non-zero. We analyze the rank of G . Firstly, the rank of G must be equal to the number of entries on the diagonal. This follows since there are at least $M - r_1$ linearly independent rows, one for each non-zero entry. So we get the following:

$$M - r_1 \leq \text{Rank}(G)$$

Secondly, we know that the final equation turns $\tilde{g}_i(x, z)$ into an equation just over z and $\tilde{h}_i(x, y)$ just over y . Because of this, we can rewrite it as $r_2 + r_3$ terms in the form of $f'_i(y)g'_i(z)$, which implies that:

$$\text{Rank}(G) \leq r_2 + r_3$$

. This equality gives us:

$$M - r_1 \leq \text{Rank}(G) \leq r_2 + r_3$$

$$M - r_1 \leq r_2 + r_3$$

$$M \leq r_1 + r_2 + r_3$$

And since the assumption was that $r_1 + r_2 + r_3$ was at most $M - 1$, we get a contradiction, proving that the identity tensor must have slice rank at least M . $\square$

2.1.2 Bounding Slice of Subtensor

We show the following:

Lemma 2.2. The if tensor T has slice rank M , then any subtensor has slice rank at most M .

Proof. The proof is very elementary, any decomposition satisfying:

$$\begin{aligned} T(x, y, z) = \sum_{i=1}^{r_1} f_i(x) \tilde{f}_i(y, z) + \sum_{i=1}^{r_2} g_i(y) \tilde{g}_i(x, z) \\ + \sum_{i=1}^{r_3} h_i(z) \tilde{h}_i(x, y) \end{aligned}$$

also holds for $(x, y, z) \in S \times S \times S$ $\square$

2.1.3 Upperbound on $SRank(CSum)$

We prove the following:

Lemma 2.3. For the function CheckSum, where:

$$CS(x, y, z) = \begin{cases} 1 & \text{if } x + y + z = 0, \quad x, y, z \in \mathbb{F}_3^n \\ 0 & \text{else.} \end{cases}$$

the slice rank of CS is $\leq 3^{cn}$ ($c < 1$)

Proof. First, the input the the checksum function is

$$(x, y, z) \in \mathbb{F}_3^n \times \mathbb{F}_3^n \times \mathbb{F}_3^n$$

We want to think of it more so as:

$$CS((x_1, \dots, x_n), (y_1, \dots, y_n), (z_1, \dots, z_n))$$

Where in every coordinate we need to have $x_i + y_i + z_i = 0$. We rewrite it as:

$$CS(x, y, z) = \prod_{i=1 \dots n} \mathbb{1}(x_i + y_i + z_i = 0)$$

If $x_i + y_i + z_i$ does not equal 0, it is either 1 or 2. Consider the following cases where we square $x_i + y_i + z_i$. If the sum mod 3 is 0, squaring it results in 0 still. However, if it is not zero, squaring both 1 and 2 yields 1. We can therefore rewrite the above equation as:

$$CS(x, y, z) = \prod_{i=1 \dots n} 1 - (x_i + y_i + z_i)^2$$

Therefore, this representation write CS as a polynomial of degree $2n$.

(Aside About Polynomials) Since I do not know about polynomials, I needed to look up separate definitions to follow here. The degree of a polynomial is the highest sum of the degrees of any single monomial. So when we square $x_i + y_i + z_i$, we get a bunch of terms of degree 2 like x_i^2 and $x_i \cdot y_i$. We multiply these terms across all $i = 1, \dots, n$. So it total the final polynomial has terms that have degree at most $2n$.

We show that any polynomial of degree $2n$ has slice rank less than full. We expand the compact representation to get:

$$\sum C_{i, \bar{j}, \bar{k}} x_1^{i_1} \cdot x_2^{i_2} \cdot \dots \cdot z_{n-1}^{k_{n-1}} \cdot z_n^{k_n}$$

We are going to use the following shorthand to simplify the above:

$$\sum C_{\bar{i}, \bar{j}, \bar{k}} x^{\bar{i}} y^{\bar{j}} z^{\bar{k}}$$

Since the total degree of any term is at most $2n$ in total and each variable has degree ≤ 2 . We group these terms into three categories:

- Terms where x 's degree $\leq \frac{2n}{3}$
- Terms where y 's degree $\leq \frac{2n}{3}$
- Terms where z 's degree $\leq \frac{2n}{3}$

Since they are not disjoint, any term not in the first gets pushed to the second and so on. Consider all terms where x 's degree is $\leq \frac{2n}{3}$:

$$C_{\bar{i}, \bar{j}, \bar{k}} x^{\bar{i}} \cdot y^{\bar{j}} \cdot z^{\bar{k}}$$

We group the each term based on the x monomial. For example, we group all terms that have the x monomial $x_1^1 x_2^2 x_3^2$. We get:

$$\sum_i x^{\bar{i}} \cdot \left(\sum_{\hat{i}, \hat{j}, \hat{k}} C_{\bar{j}, \bar{k}} y^{\bar{j}} \cdot z^{\bar{k}} \right)$$

Observe that each of these is a slice in the form $f_i(x) \cdot \tilde{f}_i(y, z)$. Therefore, we can bound r_1 as the number of ways we can sum the degrees of $x_1, \dots, x_n$ to at most $2n/3$. This part is not proven in lecture, however, he uses the following:

Lemma 2.4. The number of ways that you can form tuples $(i_1, i_2, \dots, i_n)$ such that $i_1 + i_2 + \dots + i_n \leq \frac{2n}{3}$ and every $0 \leq i_k \leq 2$ is less than or equal to $(2.757)^n$.

This implies that $r_1 \leq (2.757)^n$. The proof can extend to also show that r_2, r_3 have the same upper bound which implies:

$$SRank(CS) \leq 3 \cdot 2.757^n$$

Note that the beauty in this proof is not only getting a clean closed form solution to the check sum protocol, but beyond that having the exponents of every variable be either 0, 1, 2 which are exactly the digits of $\mathbb{R}_3$ which allows for this clean translation into slice rank. $\square$